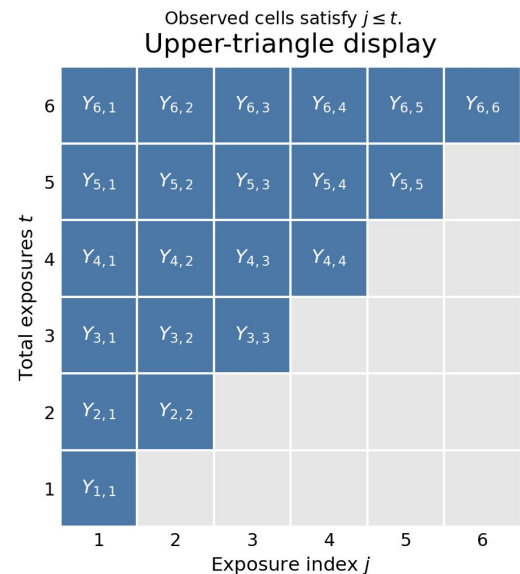


A/B Testing with Exposure-Dependent User Response

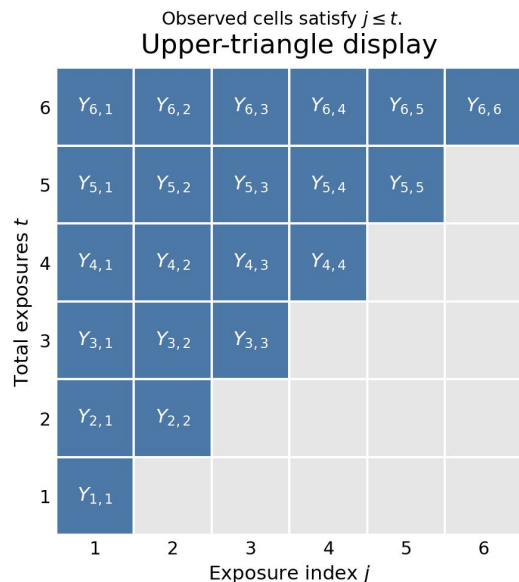
How does a **feature's effect** change with **repeated exposure**, and how long an exposure trajectory can we estimate reliably?

Scenario:

- introduce a new feature
- experiment window for 7 days
- observation: different users interacts with the feature a different number of times
- question: how does user interaction evolve over repeated interaction

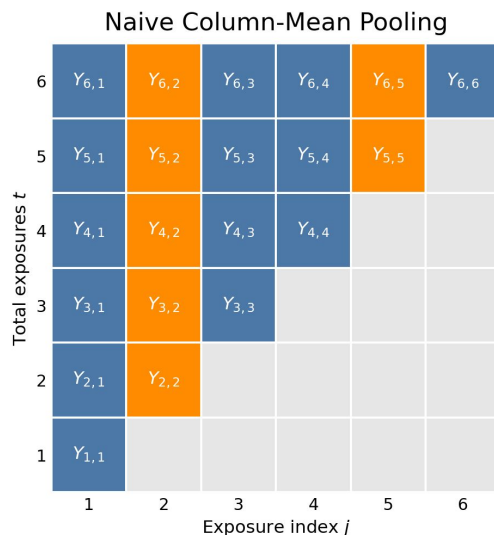


Individual "cells"



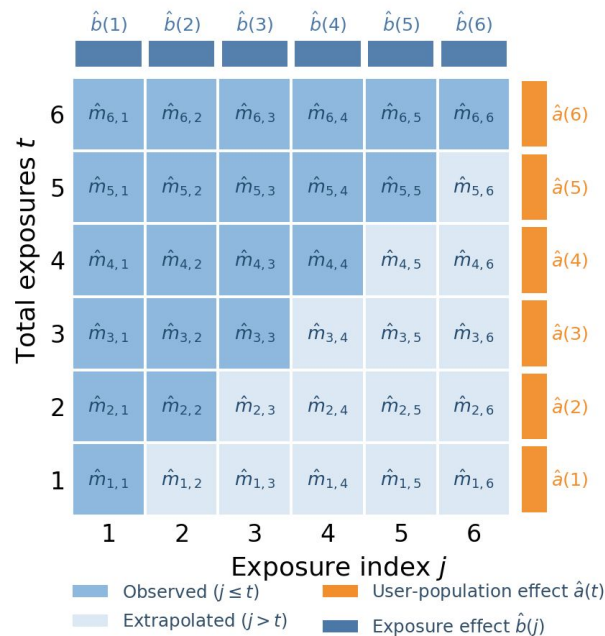
$O(B^2)$ parameters, where B denotes the maximum exposure count

Pooling by exposure



Potentially **biased** when user-response differs by total-exposure-count

Additive Fill: $\hat{m}_{t,j} = \hat{a}(t) + \hat{b}(j)$ for all cells

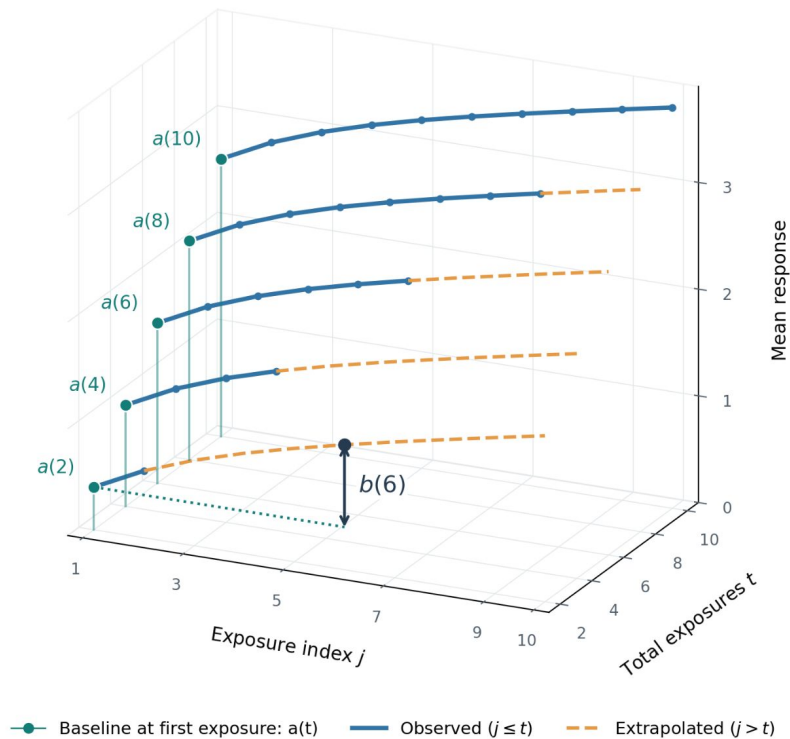


Parallel curve assumption to improve statistical efficiency
Allow different total-exposure groups to have different baseline

Different baselines, common exposure trend

$$m_{t,j} = a(t) + b(j), \quad b(1) = 0$$

$b(j)$: change from the first exposure, shared across groups

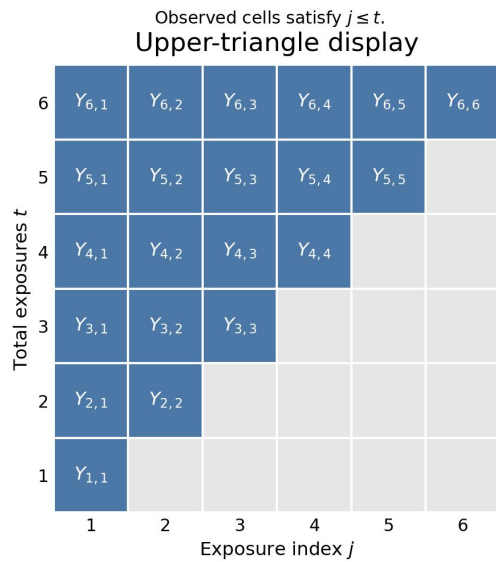


Illustrative model; baseline separation increased for clarity.

Challenge: parameters scales linearly in **maximum exposure count** (denoted by **B**)

Assume:

- 1) **monotonicity** of the exposure curve
- 2) the exposure-curve consists of a **fixed number of jumps** (denote by **k**) that does not scale with **B**



Challenge: parameters scales quadratically in **maximum exposure count** (denoted by **B**)

Assume:

- 1) **monotonicity** of the exposure curve
- 2) the exposure-curve consists of a **fixed number of jumps** (denote by k) that does not scale with B

Method:

$$\hat{\beta}_M = \Pi_{\mathcal{C}}^H(\hat{\beta}_U) = \operatorname{argmin}_{\beta \in \mathcal{C}} \|\beta - \hat{\beta}_U\|_H^2$$

$$\mathcal{C} : \text{monotone cone,} \quad \|v\|_H^2 = v^\top H v.$$

Mean-squared-error upperbound:

$$R_M = \frac{1}{B} \mathbb{E} [\|\hat{\beta}_M - \beta_0\|^2 \mid \mathcal{F}_T] \leq \sigma^2 k \log\left(\frac{eB}{k}\right) \left(\sum_{t=2}^B \frac{n_t}{N_t}\right) \left[\frac{1}{BN_B} + \sum_{j=2}^B \frac{1}{N_j j(j-1)}\right]$$

Under the assumption of $N_t \asymp n t^{-\alpha}$,

Exposure decay	Monotone risk upper bound
$0 < \alpha < 1$	$R_M \lesssim \sigma^2 \log^2 B/n$
$\alpha = 1$	$R_M \lesssim \sigma^2 \log^3 B/n$
$\alpha > 1$	$R_M \lesssim \sigma^2 B^{\alpha-1} \log^2 B/n$